

THE UNIVERSITY *of* EDINBURGH

Statistical Case Studies 2026/27 Semester 1

Project 1, part 2: Uncertainty in river flow hydrographs for flood inundation modelling

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1 Executive Summary

2 Data and Exploratory Data Analysis

- Use the tidy data from part 1
- Create hydrographs for the extreme events
- Observations of the hydrographs and understand the variety in flood event shapes

3 Statistical methodology

- fitting each of the statistical methods; The Polynomial regression, PCA, Wavelets
- uncertainty modelling for each method

4 Results and validation

- Compare the different methods
- Examine residual plots to investigate whether the method performs poorly on unusual flood events
- e.g do testing and training data to assess how well the methods peform

5 Recommendation

- Recommend the best method and justify
- explain the limitations and uncertainity with each method

6 Emulator application

- Give a brief explanation of how the emulator would be used
- e.g, explain the two seperate phases,

First, SEPA could select different parameter combinations, reconstruct corresponding hydrographs and run these through the actual flood inundation model. The resulting input-output pairs would be used to train an emulator.

Second, SEPA could repeatedly sample parameters from your fitted uncertainty distribution and feed those parameters into the trained emulator to obtain a distribution of predicted flood outcomes.

7 Conclusion